



ST. XAVIER'S COLLEGE

MAITIGHAR, KATHMANDU

THEORY OF COMPUTATION PRACTICAL INDEX SHEET

B.Sc. CSIT 4th Semester

T.U. Regd. No.:..... Class Roll No.:.....

Name of Student:..... Year/Sem.:.....

S.No.	Title of the Experiment	Submission Due	Signature	Remarks
1. String Comparison and Matching				
1.1	Write a program for identifying string over some set of symbol Set			
1.2	Write a program that takes two inputs first a string (w) and second string (s) over some alphabet set, check whether the string s is prefix, proper prefix of w or not.			
1.3	WAP that takes two inputs first a string (w) and second string (s) over some alphabet set, check whether the string s is suffix, proper suffix or not.			
1.4	WAP that takes two inputs first a string (w) and second string (s) over some alphabet set , check whether the string s is substring, proper substring or not.			
2. Finite State Automata				
2.1	Implement DFA over {a,b} that accepts {ab,bb}.			
2.2	Write a program for DFA that accepts only even number of a's and b's			
2.3	Write a program for DFA that accepts all the strings ending with 3 consecutive a's over {a,b}.			
2.4	Implement NFA over (0,1) that accepts all strings ending with 01.			
2.5	WAP for NFA over {a, b} that accepts strings having aa as substring.			
2.6	Write program to check valid C-identifiers.			

3. Regular Expression and Context Free Grammar

3.1	To implement a program that identifies valid mac address.			
3.2	To implement a program that identifies valid gmail address.			
3.3	To implement a R.E. that identifies the structure of the input: $ab^*b + aa^*b$ over $\{a, b\}$.			

4. Pushdown Automata

4.1	To implement PDA $\{a, b\}$ having equal no. Of a's and b's.			
4.2	To implement PDA over $\{a, b\}$ having number of a's followed by twice of b's			

5. Turing Machines

5.1	To implement TM over $\{a, b\}$ that recognizes string of languages $L = \{a^N b^N c^N \mid N \geq 1\}$			
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